

5 - Chemical energetics

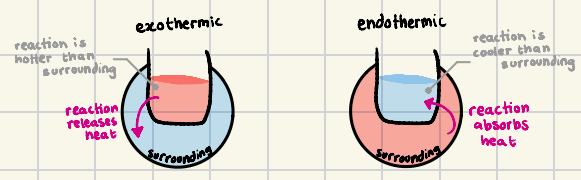
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Enthalpy change (ΔH):

→ the energy exchange between a chemical reaction and its surroundings at constant pressure

→ if energy is lost, ΔH is negative & if energy is gained, ΔH is positive

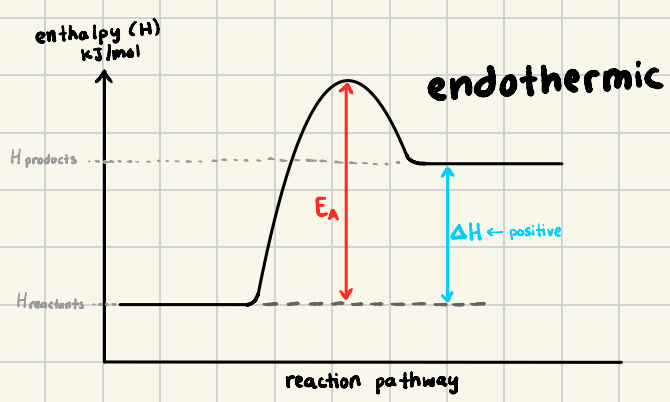
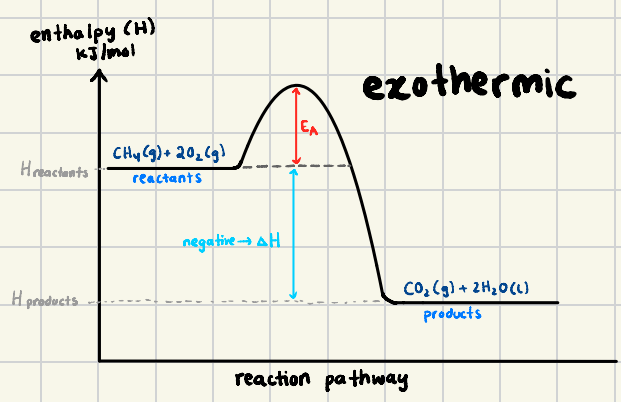
exothermic reaction endothermic reaction



$$\Delta H = H_{\text{products}} - H_{\text{reactants}}$$

enthalpy change = enthalpy of products - enthalpy of reactants

E_A : minimum amount of energy required that colliding particles must possess for a chemical reaction to occur



⇒ standard conditions [for a fair comparison between enthalpy changes, we need to use the same conditions]

- 1) pressure of 101 kPa (approximately atm)
 - 2) temperature of 293K (25°C)
- e.g. $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$

$\Delta H^\ominus = -890.3 \text{ kJ/mol}$

symbol used to indicate that an enthalpy change occurred under standard conditions

⇒ enthalpy changes of different types of reactions

- 1) enthalpy change of reaction [ΔH_r] → ΔH of a reaction that occurs exactly like its balanced chemical equation
- 2) enthalpy change of formation [ΔH_f] → ΔH when 1 mol of compound is formed from its constituent elements
- 3) enthalpy change of combustion [ΔH_c] → ΔH when 1 mol of a substance burns completely in oxygen
- 4) enthalpy change of neutralization [ΔH_{neut}] → ΔH when 1 mol of water is produced from an acid+base reaction

under standard conditions

Measuring ΔH

- using a calorimeter (polystyrene cup, vacuum flask or metal can)
- using a thermometer accurate to 0.1/0.2°C for temp. measurement

Specific Heat Capacity [c]

→ the energy required to raise the temperature of 1g of substance by 1°C (1K)

Specific Heat Capacity of water is 4.18 J/g°C

$$q = m C \Delta T$$

Annotations:
 - q : heat transferred in J
 - m : mass
 - C : specific heat capacity
 - ΔT : temperature change in °C

this is directly related to enthalpy change because 'q' represents ΔH as well

$$\Delta H = m C \Delta T \quad \text{endothermic}$$

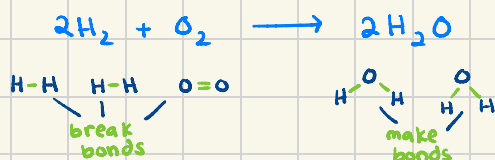
$$-\Delta H = m C \Delta T \quad \text{exothermic}$$

Bond making and breaking

→ chemical reactions involve breaking of bonds in reactants & formation of bonds in products

→ bond breaking is endothermic & bond making is exothermic

e.g

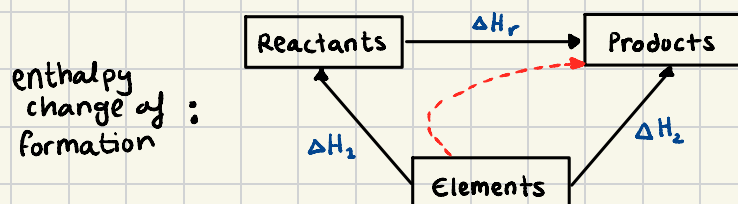


⇒ to calculate ΔH_r

- 1) identify & sum the bond energies of all bonds broken in reactant (all the energy absorbed)
- 2) identify & sum the bond energies of all bonds formed in product (all the energy released)
- 3) calculate using formula: $\Delta H_r = \underbrace{\text{Total energy absorbed}}_{\text{bonds broken}} - \underbrace{\text{Total energy released}}_{\text{bonds formed}}$

Hess's Law

→ states that the total enthalpy change for a chemical reaction is the same, regardless of the route or number of steps taken, as long as initial & final conditions are the same



$$\Delta H_2 = \Delta H_1 + \Delta H_r$$

Annotations:
 - ΔH_2 : direct route
 - $\Delta H_1 + \Delta H_r$: indirect route

$$\Delta H_r = \Delta H_2 - \Delta H_1$$

e.g

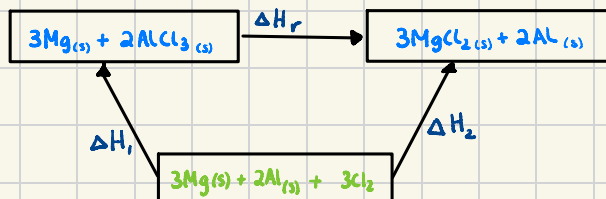
	Mg (s)	AlCl ₃ (s)	MgCl ₂ (s)	Al (s)
$\Delta H_f^\ominus$ (kJ/mol)	0	-705.6	-641.6	0

∴ calculate the total enthalpy change

$$\Delta H_2 = \Delta H_1 + \Delta H_r$$

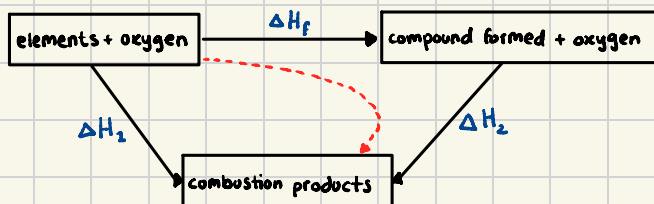
$$\Delta H_r = \Delta H_2 - \Delta H_1$$

$$\begin{aligned}
 &= 3(-641.6) + 2(0) - [3(0) + 2(-705.6)] \\
 &= (-1924.8) - (-1411.2) \\
 &= -513.6 \text{ kJ/mol}
 \end{aligned}$$

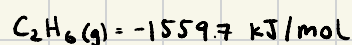
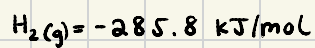
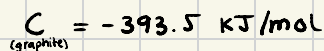
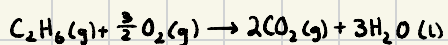
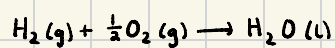
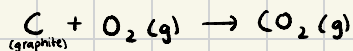


Energy level diagram showing the enthalpy change for the reaction. The reactants are at a higher energy level than the products, indicating an exothermic reaction. The energy change is labeled as -513.6 kJ/mol.

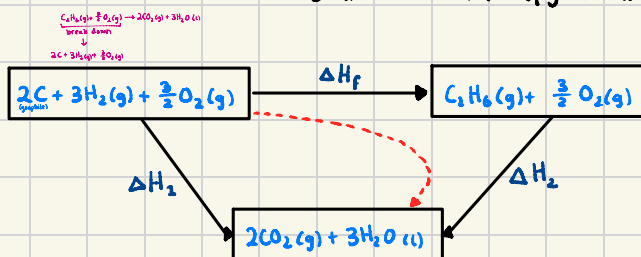
enthalpy change of :
combustion



e.g



∴ calculate the standard enthalpy change of the formation of ethane, C_2H_6



$$\Delta H_1 = \Delta H_f + \Delta H_2$$

$$\begin{aligned} \Delta H_f &= \Delta H_1 - \Delta H_2 \\ &= 2(-393.5) + 3(-285.8) - (-1559.7) \\ &= -1644.4 + 1559.7 \\ &= -84.7 \text{ kJ/mol} \quad \checkmark \end{aligned}$$